

YEABSIRA HAWAZ

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EDUCATION

Massachusetts Institute of Technology

M.S. in Electrical Engineering and Computer Science

B.S. in Electrical Engineering and Computer Science;

Cambridge, MA

Anticipated 2027

Class of 2026

Coursework: Grad Electromagnetics, Grad Digital Systems/RFSoc, Accelerated GPU Programming, Digital Systems Lab, Semiconductors, Microcomputer Project Lab, Constructive Computer Architecture GPA: 4.5/5.0

EXPERIENCE

MIT Kavli Institute for Astrophysics and Space Research

Graduate Researcher

Cambridge, MA

09/2026 – Present

- Writing System Verilog for FPGA-Based X-ray Telescope CCD readout: ADC capture, MCDS, and filtering
- Designing a single-ended to differential signal conversion PCB to validate high-speed signal integrity between CCD front-end and FPGA

CRABI Robotics

Hardware Engineering Intern

Cambridge, MA

06/2026 – 09/2026

- Developed a FPGA-based underwater ultrasonic data acquisition system on an Artix-7, building a full ADC→FIR-Filter→DDR3→UDP Ethernet streaming pipeline for hydrophone signal capture.
- Designed a FPGA-based Ultrasonic Imaging system for subsurface imaging of ship-hulls. Implemented matched filtering, and peak detection on a Artix-7 FPGA to detect TOF of Ultrasonic Pulses at 70V
- Designed a size-constrained 4-layer PCB with a Ethernet PHY, including differential pair routing, for high speed comms with a Artix-7 in a waterproof enclosure

zeroRISC

Hardware Engineering Intern

Boston, MA

01/2026 – 02/2026

- Developed an AXI host interface for the Ibex RISC-V core, replacing its native TileLink-UL connection to the OpenTitan Chip's crossbar with an AXI4 master port

Hardware Engineering Intern

06/2025 – 08/2025

- Developed SystemVerilog implementations of a TileLink-UL to AXI4 bridge and an AXI CSR interface bridge, enabling the OpenTitan chip to interface with peripherals via AXI
- Wrote Verilator C++ test benches and validated designs on a Xilinx VCU118 FPGA board with various Xilinx AXI IP cores, and verified against existing tape-out UVM DV

MIT Computer Science & Artificial Intelligence Laboratory

Undergraduate Researcher, Computer Architecture Group

Cambridge, MA

01/2024-Present

- Architected and implemented a 2D-torus Network-on-Chip for a sparse linear algebra accelerator, achieving 400 MHz synthesis on an AMD Alveo U55C
- Implemented dual virtual channels and a two-stage pipeline, reducing average packet latency to 17 cycles—1.3× faster than comparable mesh NoCs under heavy traffic.
- Validated the design via cycle-accurate simulation and hardware integration, passing stress-tests with 16×16 networks under 150K+ message workloads.

PROJECTS

RadioHead: Real-Time FM Radio on RFSoc

09/2025-12/2025

- Built a real-time FM demodulator on a Xilinx RFSoc 4x2, implementing a complex-conjugate demod, 64-bit CORDIC, and FIR filtering entirely in SystemVerilog
- Decoded RDS metadata by isolating the 19 kHz pilot tone, generating its third harmonic via clipping, and demodulating the 57 kHz BPSK subcarrier with zero-crossing symbol synchronization

Holoforge: A Gesture Controlled 3D Model Viewer on Spartan-7

10/2024-06/2025

- Designed and implemented a camera-controlled 3D model viewer on a Xilinx FPGA, with a custom graphics pipeline (pre-proc, shader, rasterizer) and AXI-backed DDR3 framebuffer
- Developed centroid tracking using an Adafruit camera to control scene view in real time
- Built a custom AXI stacker to optimize out-of-order writes to DDR3 via the MIG, enabling double-buffered rendering at 360×180 resolution

Jazzy Dude: Real-Time Digital Instrument Synthesizer on PSoC

03/2025-05/2025

- Designed and implemented a jazz-based instrument synthesizer on a PSoC 5LP in C, generating saxophone and piano sounds using custom ADSR envelopes and phase accumulators.
- Programmed real-time playback and transposition of pre-programmed jazz standards, enabling dynamic key and tempo adjustments entirely in firmware.

TECHNICAL SKILLS

Languages: SystemVerilog, Bluespec, C/C++, CUDA, RISC-V Assembly, Python, x86 Assembly

Developer Tools: Xilinx Vivado, KiCad, Verdi, Verilator, Vivado ILA, TCL, GIT, Linux, Bash Scripting